Supplementary Data

Table S1: Rheological, Mechanical, Thermal, and Structural Parameters Across Reviewed Literature

Paper & Citation	Primary Matrix / Genotypes Studied	Moisture / Dry Matter / Preparation	Storage Modulus (G′ or E′)	Loss Modulus (G″ or E″)	Loss Factor (tanδ)	Yield Stress (τ0 or σc)	Consistency Index (K)	Flow Behavior Index (n)	TPA Hardness / Extrusion Force	TPA Adhesiveness / Tack	TPA Springiness & Cohesiveness	Specific Mechanical Energy (SME) / Shear Degradation	Thermal / DSC Parameters (To,Tp,Tc,ΔH)	Starch & Amylose Fractions
Akissoé et al. (2011)	*Dioscorea rotundata* (Anago, Gnidou, Kokoro, Laboco, Morokoro) & *D. alata* (Florido)	Raw DM: 24.5–48.6 g/100g wb; Cooked: boiled 15 min, adjusted to 25 g/100g wb	DMA Flexural E' (1 Hz): Raw: 77--94 × 10^4 Pa; Max: 78--94 × 10^4 Pa; Cooked: 26--30 × 10^4 Pa	DMA Flexural E'': Raw: 24--27 × 10^4 Pa; Max: 57--67 × 10^4 Pa; Cooked: 19--23 × 10^4 Pa	No clear thermal transition during DMA cooking run	—	—	—	Extrusion Firmness (N/cm^2): Anago: 5.1±0.7, Florido: 2.8±0.7, Gnidou: 4.4±0.7, Kokoro: 4.6±0.7, Laboco: 2.6±0.7, Morokoro: 2.1±0.7	Chen-Hoseney Tack (A_{neg}): Anago: 1.7±0.3 N · s, Florido: 0.9±0.3 N · s, Laboco: 0.6±0.3 N · s	Adhesion Distance: 1.1–3.2 mm; Two-cycle TPA springiness ratio: —	Destructuration model: Firmness = $-6.18 + 0.41\,\text{DM} - 0.14\,\text{St} + 0.45\,\text{Amy}$ ($R^2 = 0.75$)	T_o = 66.6--72, T_p = 66.6--75, ΔH = 9.0--12.8	Soluble starch: 22.3–46.6 g/100g db; Soluble amylose: 5.5–13.1 g/100g db
Lee & Yoo (2011)	*Ipomoea batatas* (Sweet potato starch: native & hydroxypropylated MS 0.042–0.153)	Dispersions: 5% w/v (95 °C, 30 min for flow/dynamic); 10% w/v for gel strength	Oscillatory Shear G' (25 °C, 6.28 rad/s): Native: 29.5±0.3 Pa; MS 0.042: 19.5±0.4 Pa; MS 0.153: 10.7±0.1 Pa	Oscillatory Shear G'' (25 °C, 6.28 rad/s): Native: 11.0±0.0 Pa; MS 0.042: 8.54±0.28 Pa; MS 0.153: 5.51±0.02 Pa	$\tan\delta$ (G''/G'): Native: 0.37±0.01; MS 0.042: 0.44±0.01; MS 0.153: 0.51±0.01	Casson Yield (σ_{0c}): Native: 38.0±1.3 Pa; MS 0.042: 30.6±0.7 Pa; MS 0.153: 20.6	Power-law K (Pa · s^n): Native: 20.4±0.5; MS 0.042: 15.8±0.4; MS 0.153: 9.57±0.3	Power-law n: Native: 0.38±0.01; MS 0.042: 0.41±0.01; MS 0.153: 0.47±0.01	Brookfield Gel Strength (g): Native: 1054±67.2 g; MS 0.042: 1048±42.5 g; MS 0.153: 224±7.1 g	—	—	Viscosity drop under shear attributed to mechanical scission of starch networks	Native: T_o = 66.6 ± 0.2 °C, T_p = 76.9 ± 0.2 °C, T_c = 82.6 ± 0.2 °C, ΔH = 16.5 ± 0.2 J/g	Native amylose: 24.6 g/100g

						±0.4 Pa								
Ayetigbo et al. (2025)	*D. rotundata* (8 genotypes) & *D. alata* (11 genotypes)	Steamed in QZP/6000 yam pounder (22 min), rotary pounded 1–3 min	Bi-extensional Viscosity (BEV): *D. rotundata*: $8.98 \pm 2.63 \times 10^4$ Pa·s; *D. alata*: $7.34 \pm 3.50 \times 10^4$ Pa·s	—	—	—	—	—	Extensogram Peak Force (EPF, N): *D. rotundata*: 0.62±0.11 N (Krenglè: 0.78, TDr1443002: 0.80); *D. alata*: 0.29±0.10 N	—	Extensibility (EXT, mm): *D. rotundata*: 13.3±4.1 mm; *D. alata*: 5.9±1.8 mm; Work (EA, N·mm): 4.6±2.02 vs 1.1±0.5	EXT = $-9.587 - 0.778$ SOL 0.0133 PV ($R^2 = 0.828, P = 0.0051$)	Pasting Temp: 75.9--79 (*D. rotundata*) vs 77.6--81 (*D. alata*)	Amylose (DSC): *D. rotundata*: 23.9±3.9%; *D. alata*: 23.1±1.9%; Pectin: 771–1147 mg/100 g
Afolabi et al. (2025)	*D. rotundata* Poir. (Abuja, Lagos, Pampas) across 3 mechanical pounding drives	Boiled cubes 20 min, processed 5 min (Mortar, Machine, Attrition mill)	—	—	—	—	—	—	Instrumental Hardness (N): Mortar: 5.02--5.; Machine: 2.15--2.; Attrition mill: 1.64--2.	Adhesiveness (g·s^{-1}): Mortar: -78.29 to -86.18; Machine: -29.28 to -33.17; Attrition mill: -209.78 to -226.67	Springiness (mm): Mortar: 0.29–0.33; Machine: 0.48–0.56; Attrition: 0.54–0.63; Cohesiveness: Mortar: 0.70; Attrition: 0.29–0.34	Attrition over-shearing causes 60% hardness collapse, >650% adhesiveness surge, and 55% cohesiveness loss	RVA Pasting Temp: 82.85 ± 0.71 °C to 84.10 ± 0.07 °C	Starch solubility: 3.50±0.16% to 4.76±0.06%; Swelling: 10.36–12.81 g/g
Gumus & Wauton (2014)	*Dioscorea rotundata* starch paste (0.83% borax blend)	Dissolved dry starch (30 g/250 mL 0.01M HCl), gel heated, blended	—	—	—	Casson Yield (σ_c): 40 °C: 5.14 Pa; 50 °C: 5.043 Pa; 60 °C:	Power-Law K (Pa·s^n): 40 °C: 26.36; 50 °C: 25.47; 60 °C: 24.77 ($E_a =$ 2.69 kJ/)	Power-Law n: 40 °C: 0.388; 50 °C: 0.388; 60 °C: 0.336; Constitutive eq: $\tau =$ $26.36\gamma^{}$	—	—	—	Continuous shear (1--20 s^{-1}) causes viscosity collapse from 23×10^3 cP to 3×10^3 cP	—	—

						4.978 Pa								
Adeboyejo (PhD Thesis)	*Dioscorea dumetorum* (White/Yellow bitter yam) & *Colocasia esculenta* / *Xanthosoma*	Root DM: White (39.11%), Yellow (39.55%), Cocoyam (26.24–37.00%); Starch moisture: 10.32–12.90%	— (Conceptually defined)	— (Conceptually defined)	— (Conceptually defined)	—	—	—	Gel resistance defined conceptually; Least Gelation Concentration: 4–14%	—	—	Acid thinning induces chain scission, eroding growth rings; RVA peak drops from >366 to <4 RVU	White bitter: $T_o =$ 88.67 °C, $T_p =$ 99.90 °C, $T_c =$ 113.85 °C, $\Delta H =$ 16.88 J/g; Yellow: $T_o =$ 77.51 °C, $\Delta H =$ 9.26 J/g	Amylose: White bitter (8.69±0.11%), Yellow (17.95±0.16%), *C. esculenta* (10.11–24.71%)
Adesokan et al. (2024)	*D. alata* (58 lines) & *D. rotundata* (7 lines incl. Meccakusa, Ojuiyawo)	Tuber DM: Water yam (29.85 ± 4.00 g/100); White yam (32.43 ± 3.80 g/100)	—	—	—	—	—	—	Ottawa Cell Extrusion Hardness (g): Range: 7,440.72–25,056.47 g; Mean: 12,554.2 4,447.58	Instrumental adhesiveness: -70 to -10 g · s; Plotted in matrix ($r =$ 0.50 with sensory stickiness)	Extrusion Work: 117,569 45,779.4 s; Chewiness: 10,000–50,000 g	Cooking time (9–15.5 min) and water absorption inversely govern shear extrusion resistance	Boiling cooking time: 11.17 ± 1.85 min (range: 9.0–15.5 min)	Starch: 42.81–83.81 g/100g; Amylose: 29.27–38.52 g/100g; Sugar: 4.46–6.95 g/100g
Argaw et al. (2024)	*D. bulbifera* (SC4), *D. cayenensis* (A), *D. rotundata* (AC-04), *D. cayenensis* complex (CH-21)	8% w/v freeze-dried flour in water, gelatinized at 95 °C for 1 h, tested at 25 °C	Oscillatory Shear G' (25 °C, 0.2% strain, 1–10 Hz): *D. bulbifera*: 4900–6200 Pa; *D. cayenensis* complex: 4500–5250 Pa; *D. cayenensis*: 1650–2150 Pa; *D. rotund*	Oscillatory Shear G'' (1–10 Hz): *D. bulbifera*: 610–780 Pa; *D. cayenensis* complex: 250–405 Pa; *D. cayenensis*: 170–230 Pa; *D. rotundata*:	Strictly $\tan\delta <$ 1.0 across 0.1–10 Hz ($G' > G''$); *D. rotundata*: $\tan\delta \approx$ 0.08--0.	—	—	Evaluated via frequency sweep shear thinning; steady-shear power-law index: —	—	—	—	Low RVA breakdown in *D. rotundata* (163 cP) signals structural resistance against over-softening	*D. rotundata*: $T_o =$ 47.23 ± 1.89 °C, $T_p =$ 71.70 °C, $T_c =$ 105.20 ± 5.55 °C, $\Delta H =$ 2.30 ± 0.27 J/g	Total Starch: 52.97–64.63%; Amylose: *D. cayenensis* (23.53 %), *D. rotundata* (19.85 %), *D. bulbifera* (16.92 %)

			ata: 1450–1980 Pa	115–220 Pa										
Correa Mosquera et al. (2025)	Native Yam (*Dioscorea* sp.) starch in baked dough matrices (pandebono, pandeyuca, cookies, panderos)	Starch extracted via water maceration, sieved (80 µm), dried at 50 °C (12 h); baked at 170–210 °C	—	—	—	—	—	—	TA-XT2i Hardness (N): Pandebono: 17,352 ±4,530; Pandeyuca: 15,385 ±1,463; Cookies: 5,193± 610; Panderos: 1,574± 608	Adhesiveness (g): Pandebono: -0.98±0.58; Pandeyuca: -5.07±0.59 g	Pandebono: Springiness 0.80±0.08, Cohesiveness 0.51±0.10, Chewiness 7,423± 3,946 N	Higher gelatinization onset (84.2 °C) causes internal ungelatinized core fractures under rapid baking	Stated gelatinization temperature: 84.2 °C (yam starch) vs 64.0 °C (cassava starch)	Starch Amylose Content: Yam (24.08 %) vs Cassava (17.0%)
Lozano et al. (2022)	*Dioscorea rotundata* tuber mucilage in hydrocolloid systems (0.4% and 0.8% CS with CMC)	Pilot bubble-flotation extraction (1:8 ratio), freeze-dried (-50 °C, 60 h)	Power-law oscillatory ($G' = K'\omega^{n'}$): 100% Yam (0.4% CS): $K' =$ $1.589 \pm$ 0.008 P $s^{n'}$; (0.8% CS): $K' =$ $1.629 \pm$ 0.123	Power-law oscillatory ($G'' = K''\omega^{n'}$): 100% Yam (0.4% CS): $K'' =$ $0.369 \pm$ 0.021 P $s^{n'}$; (0.8% CS): $K'' =$ $0.907 \pm$ 0.084	100% Yam (0.4% CS): $\tan\delta =$ $1.08 \pm$ 0.11; (0.8% CS): $0.75 \pm$ 0.06; 80:20 (Yam: CMC): $0.36 \pm$ 0.01	—	Ostwald-de Waele K (Pa · s^n): 100% Yam (0.4% CS): $0.011 \pm$ 0.001; (0.8% CS): $0.023 \pm$ 0.001; 20:80 (Yam: CMC): $6.659 \pm$ 1.522	Power-law n: 100% Yam (0.4% CS): $0.955 \pm$ 0.022; (0.8% CS): $0.900 \pm$ 0.001; 20:80 (Yam: CMC): $0.471 \pm$ 0.046	Hardness in N: —; Evaluated physical stability: Overrun (56.27 – 78.27 %), Melting time (33.33 –70.00 min)	—	—	Pseudoplastic shear thinning with zero time-dependent hysteresis between upward/downward shear loops	Freezing point depression ranged from $-6.10 \pm$ 0.55 °C to $-2.80 \pm$ 0.04 °C	Protein-polysaccharide complexation
Otegbayo et al. (2024)	16 accessions across 4 *Dioscorea* species (*D. rotundata*, *D. alata*, *D. bulbifera*, *D.*	8% db purified starch gel, gelatinized at 95 °C for 15 min, tested at 25 °C	Oscillatory Shear G' (0.1–100 rad/s): *D. bulbifera*: 700–3,400 Pa; *D.*	Oscillatory Shear G'' (0.1–100 rad/s): *D. bulbifera*: 30–155 Pa; *D.*	Strictly $\tan\delta <$ 1.0; *D. rotundata* highest ($\tan\delta =$ 0.20--0., up to 0.65 at	—	Complex viscosity (η^*): *D. bulbifera*: 2,000 – 10,000 Pa · s; *D.*	Pseudoplastic shear thinning across all 16 varieties ($\partial\eta^*/\partial\omega <$ 0)	—	—	—	Unmetered shear on low complex viscosity *D. rotundata* gels ($\eta^* <$ 600 Pa · s) drives cohesive loss	**D. rotundata:** $T_o =$ $71.70 \pm$ 0.77 °C , $T_p =$ $76.32 \pm$ 1.36 °C , $T_c =$ $80.36 \pm$ 1.24 °C , $\Delta H =$	Amylose: *D. bulbifera* (25.31 %), *D. alata* (24.44 %), *D. rotundata* (24.26 %), *D.*

	dumetorum)		*dumetorum*: 200–520 Pa; *D. alata*: 100–880 Pa; *D. rotundata*: 50–380 Pa	*alata*: 10–130 Pa; *D. dumetorum*: 20–105 Pa; *D. rotundata*: 15–90 Pa	low ω); *D. bulbifera* lowest (0.03–0.11)		*alata*: 400–850 Pa · s; *D. rotundata*: 250–600 Pa · s						17.67 ± 0.61 J/g	*dumetorum* (17.08 %)
Arueya & Ojesanmi (2019)	Native & modified *D. alata* starch (cv. Dagidagi)	Hydroxypropylated starch powder / 8% paste	— (RVA Peak Viscosity = 583.6 RVU)	—	—	—	—	—	—	—	Least Gelation Concentration (LGC): 6–10%	Chemical hydroxypropylation disrupts inter-chain hydrogen bonds, altering paste viscosity	RVA Pasting Temp: 84.0 °C (Native) down to 61.9 °C (highest modification)	Amylose Content: Native *D. alata* = 28.30± 0.42%

Table S2: Structural, Microscopic, Molecular, and Empirical Quality Parameters Across Reviewed Literature

Paper & Citation	Starch Granule Morphology & Particle Size (μm)	Crystallinity % & XRD Polymorph Type	Extracellular Matrix & Pectin Content	Cell Ghost Integrity & Free-Starch Ratio	Swelling Power (g/g) & Solubility (%)	Pasting Temperatures & Viscosity (RVA/AMCL)	Correlation Coefficients (r) / Regression (R2)	Primary Structural Mechanism / Finding
Akissoé et al. (2011)	Oval and polyhedral granules; mean size range: 15--35 μm	B-type crystallinity; percentage unquantified	Water-soluble pectin content: 0.8–2.3% dry basis	Cell integrity retained in raw tuber; cellular rupture releases amylose into continuous serum	Swelling power at 85 °C: 11.2–18.5 g/g; Solubility: 8.5–16.2%	Pasting Temp: 68.5–76.2 °C; Peak Viscosity: 450–820 RVU; Breakdown: 120–310 RVU	Firmness $= -6.18 + 0.41\,\mathrm{DM} - 0.14\,\mathrm{St} + 0.45\,\mathrm{Amy}$ ($R^2 = 0.75$)	High amylose and dry matter increase firmness and stickiness due to starch leaching into the inter-cellular matrix during thermal processing.
Lee & Yoo (2011)	Round and polygonal; mean size: 18.2 μm (native)	A-type polymorph; Native crystallinity = 38.2%; Hydroxypropylated: 34.1–36.8%	—	Chemical substitution weakens granule structural integrity, promoting early swelling	Swelling power at 80 °C: Native = 18.5 g/g, MS 0.153 = 31.2 g/g; Solubility: 8.2% to 22.4%	Pasting Temp: 68.4 °C (Native), 52.1 °C (MS 0.153); Peak Viscosity increases with substitution level	Power-law fit: $R^2 \geq 0.98$; Casson model fit: $R^2 \geq 0.99$	Hydroxypropyl groups introduce steric hindrance, disrupting internal hydrogen bonding and lowering yield stress and elastic modulus (G').
Ayetigbo et al. (2025)	Starch granules embedded within parenchymatous cell walls	—	Pectin: 771–1147 mg/100g db across species	Cell integrity lost during mechanical pounding, releasing starch matrix	Swelling power: 12.1–16.8 g/g; Solubility: 5.2–11.4%	Pasting Temp: 75.9–79.2 °C (*D. rotundata*) vs 77.6–81.1 °C (*D. alata*)	$\mathrm{EXT} = -9.587 - 0.778\,\mathrm{SOL}_{\mathrm{flour}} + 0.0133\,\mathrm{PV}_{\mathrm{flour}}$ ($R^2 = 0.828, P = 0.0051$)	Extensibility and bi-extensional viscosity are governed by soluble starch fraction and flour peak viscosity during rotary pounding.
Afolabi et al. (2025)	Mechanical shear ruptures cellular structures	—	Cell wall integrity compromised by attrition grinding	Cell wall destruction releases free starch into inter-cellular gaps	Swelling power: 10.36–12.81 g/g; Solubility: 3.50–4.76%	Pasting Temp: 82.85–84.10 °C; Breakdown: 12.50–28.10 RVU	Strong negative correlation between shear rate and dough hardness ($r = -0.89$)	Attrition mill shear causes excessive cell wall breakdown, resulting in a 60% drop in hardness and a >650% rise in stickiness.
Gumus & Wauton (2014)	Borax-crosslinked starch network	B-type crystalline domains	—	Complete starch network dissolution under high shear	Swelling power increases with borax crosslinking	Gelatinization complete during thermal blending	Casson yield model fit ($R^2 = 0.991$); Power-law fit ($R^2 = 0.987$)	Continuous shear breaks borax-starch crosslinks, causing a rapid viscosity drop from 23,000 cP to 3,000 cP.
Adeboyejo (PhD Thesis)	Small, polygonal granules (*C. esculenta*) vs large, oval granules (*D. dumetorum*)	B-type (*D. dumetorum*) and A-type (*C. esculenta*) patterns	Pectin and mucilage content in cocoyam matrices	Acid hydrolysis strips amorphous starch rings	Swelling power: 4.2–11.8 g/g; Solubility: 2.1–18.5%	Pasting Temp: 77.5–88.7 °C; Peak Viscosity drops drastically after acid treatment	Inverse relation between acid treatment time and peak viscosity ($r = -0.94$)	Acid thinning attacks amorphous growth rings, causing RVA peak viscosity to drop below 4 RVU.

Adesokan et al. (2024)	Large oval granules in *D. rotundata*; smaller granules in *D. alata*	—	Tuber cell wall structural resistance	Cooking breaks cell wall barriers, allowing starch swelling	Swelling power: 9.5–17.2 g/g; Solubility: 6.1–14.8%	Boiling time: 9.0–15.5 min; RVA parameters variable	Ottawa cell force correlates with sensory stickiness ($r = 0.50$)	Longer cooking times soften tuber cell walls, reducing extrusion force and dough firmness.
Argaw et al. (2024)	Smooth, oval/spherical granules across species	B-type polymorph in *D. rotundata* and *D. bulbifera*	—	Freeze-drying preserves cell ghost network integrity	Swelling power: 8.8–14.2 g/g; Solubility: 4.5–9.8%	Pasting Temp: 71.7 °C (*D. rotundata*); Low breakdown (163 cP)	G' correlates positively with amylose content ($r = 0.82$)	Low RVA breakdown in *D. rotundata* indicates stable gel networks that resist thermal and mechanical degradation.
Correa Mosquera et al. (2025)	Smooth oval yam starch granules vs round cassava granules	B-type pattern (yam starch) vs A-type (cassava)	—	Un-gelatinized cores form due to high gelatinization temperature	Swelling power: 11.5 g/g (yam) vs 18.2 g/g (cassava)	Gelatinization Temp: 84.2 °C (yam) vs 64.0 °C (cassava)	$R^2 = 0.91$ for cookie hardness vs starch inclusion level	High gelatinization onset temperature in yam starch causes incomplete swelling during fast baking, increasing product hardness.
Lozano et al. (2022)	Freeze-dried mucilage mesh	Amorphous biopolymer matrix	Soluble hydrocolloid pectin-protein complex	—	High water retention capacity	Viscosity increases with concentration; no thermal hysteresis	Ostwald-de Waele fit ($R^2 \geq 0.97$)	Yam mucilage acts as a weak viscoelastic gel; blending with CMC increases pseudoplastic behavior and storage modulus (G').
Otegbayo et al. (2024)	Granule size: *D. bulbifera* (small) to *D. rotundata* (large oval)	B-type polymorph dominant across species	—	Starch network integrity dictates elastic recovery	Swelling power: 9.1–16.5 g/g; Solubility: 3.8–12.1%	Pasting Temp: 71.7–76.3 °C; Peak Viscosity: 180–420 RVU	G' positively correlates with amylose content ($r = 0.88$)	Higher amylose content in *D. bulbifera* strengthens gel networks, increasing G' compared to *D. rotundata*.
Arueya & Ojesanmi (2019)	Modified granules show surface etching	Crystalline region altered by hydroxypropylation	—	Chemical modification disrupts granule core	Swelling power increases from 8.4 to 22.1 g/g after treatment	Pasting Temp drops from 84.0 °C to 61.9 °C; Peak Viscosity: 583.6 RVU	$R^2 = 0.96$ for substitution degree vs pasting temperature	Hydroxypropylation lowers pasting temperature and increases paste clarity by weakening internal hydrogen bonds.

Table S3: Process Conditions, Sensory Correlations, Rheological Models, and Applications Across Reviewed Literature

Paper & Citation	Unit Operations & Processing Variables (Temp, Shear, Time, Moisture)	Sensory Texture Attributes & Quantitative Scores	Mathematical Models & Constitutive Equations	R2 / Model Error Metrics	Specific Mechanical Energy (SME) / Processing Thresholds	Critical Quality Factors & Processing Guidelines	Primary Food / Industrial Application
Akissoé et al. (2011)	Boiled 15 min; DM content: 24.5–48.6 g/100g wb; Moisture adjusted to 75 g/100g	Sensory Firmness: 1.2–5.8; Stickiness: 0.8–4.2 (10-point scale)	Firmness Regression: Firmness $= -6.18 + 0.41\,\mathrm{DM} - 0.14\,\mathrm{St} + 0.45\,\mathrm{Amy}$	$R^2 = 0.75$ for multi-linear regression	Processing limit: boiling > 20 min over-softens tubers	High dry matter and amylose govern firmness and stickiness in boiled/pounded yams.	Boiled yam, pounded yam (foutou), traditional yam pastes
Lee & Yoo (2011)	Chemical hydroxypropylation (MS 0.042–0.153); Heated at 95 °C for 30 min	Viscoelastic gel texture becomes softer and more spreadable at higher MS	Casson: $\sigma^{0.5} = \sigma_{0c}^{0.5} + K_c^{0.5}\gamma^{0.5}$; Power-law: $\sigma = K\gamma^n$	$R^2 \geq 0.98$ (Power-law); $R^2 \geq 0.99$ (Casson)	Mechanical shear causes network scission at shear rates $> 100\ \mathrm{s}^{-1}$	Hydroxypropylation lowers yield stress and gel strength while improving freeze-thaw stability.	Frozen desserts, modified starch thickeners, clear sauces
Ayetigbo et al. (2025)	Steamed 22 min in QZP/6000 yam pounder; Rotary pounded 1–3 min	Extensibility (sensory stretchability), Moldability, Softness	Extensibility: EXT $= -9.587 - 0.778\,\mathrm{SOL}_{\mathrm{flour}} + 0.0133\,\mathrm{PV}_{\mathrm{flour}}$	$R^2 = 0.828$ ($P = 0.0051$)	Rotary pounding $>$ 3 min over-shears dough, degrading extensibility	Soluble starch fraction and peak viscosity control stretchability and bi-extensional viscosity.	Mechanized pounded yam (iyan) manufacturing, instant yam flour
Afolabi et al. (2025)	Mortar & pestle vs Machine pounder vs Attrition mill (5 min processing)	Hardness, Stretchability, Hand-feel stickiness, Smoothness	Linear correlation models between processing drive shear and TPA metrics	$r = -0.89$ (shear rate vs hardness)	Attrition mill high shear causes 60% hardness collapse and >650% adhesiveness surge	Avoid high-shear attrition milling to prevent catastrophic structural breakdown of dough.	Mechanized pounded yam processing, industrial food equipment design
Gumus & Wauton (2014)	Borax crosslinking (0.83%); Continuous shear (1--20 s^{-1}) at 40–60 °C	Gel smoothness, spreadability, surface tack	Casson: $\tau^{0.5} = \tau_0^{0.5} + \eta_c^{0.5}\dot{\gamma}^{0.5}$; Power-law: $\tau = K\dot{\gamma}^n$	$R^2 = 0.991$ (Casson fit); $R^2 = 0.987$ (Power-law fit)	Continuous shear causes viscosity collapse from 23,000 cP to 3,000 cP	Shearing above 20 s^{-1} permanently breaks crosslinked borax-starch networks.	Technical starch gels, specialized adhesive pastes, modified starch thickeners
Adeboyejo (PhD Thesis)	Acid thinning (1.0 M HCl, 20–50 °C); Heat-moisture treatment	Gel resistance, pastiness, mouth-coating, gel firmness	RVA viscosity degradation kinetics modeling	$r = -0.94$ (acid time vs peak viscosity)	Acid thinning drops peak viscosity from >366 to <4 RVU	Acid hydrolysis dismantles starch rings, lowering pastiness while increasing solubility.	Digestible starch matrices, low-viscosity weaning foods
Adesokan et al. (2024)	Boiling (9–15.5 min) followed by Ottawa cell shear extrusion	Sensory Stickiness, Firmness, Cohesiveness, Moldability	Pearson correlation matrix (r) and principal component analysis (PCA)	$r = 0.50$ (extrusion force vs sensory stickiness)	Boiling time threshold: 11.17 ± 1.85 min for optimal extrusion force	Cooking time directly controls extrusion resistance; undercooked tubers increase hardness.	Pounded yam suitability screening, breeding selection for texture
Argaw et al. (2024)	Gelatinized at 95 °C for 1 h; Frequency	Gel firmness, elastic bounce,	Power-law Frequency: $G' =$	$R^2 \geq 0.95$ across species	Low RVA breakdown (163 cP) indicates	High G' and low breakdown make *D. rotundata* ideal for	Solid gel foods, thickeners,

	sweep (0.1--10 Hz) at 25 °C	mouthfeel smoothness	$K'\omega^{n'}$; $G'' = K''\omega^{n''}$		resilience to shear processing	firm, elastic gel products.	stable food pastes
Correa Mosquera et al. (2025)	High-temperature short-time baking (170--210 ˚C)	Hardness, Crunchiness, Fracturability, Mouthfeel chewiness	Linear regression for starch substitution vs product hardness	$R^2 = 0.91$ for cookie hardness	Thermal baking threshold: 170 °C required for complete gelatinization	High gelatinization temperature (84.2 ˚C) requires controlled moisture to avoid brittle cores.	Gluten-free bakery goods (pandebono, cookies, panderos)
Lozano et al. (2022)	Bubble-flotation extraction; Freeze-drying (-50 °C, 60 h); Blending with CMC	Emulsion stability, overrun, mouthfeel creaminess	Ostwald-de Waele: $\tau = K\dot{\gamma}^n$; Oscillatory: $G' = K'\omega^{n'}$	$R^2 \geq 0.97$ across formulations	Over-mixing does not induce time-dependent hysteresis	Blending yam mucilage with CMC increases elastic modulus (G') and pseudoplasticity.	Ice cream stabilizers, foam stabilization, food emulsions
Otegbayo et al. (2024)	Gelatinized at 95 °C for 15 min; Tested at 25 °C ($\omega =$ 0.1--100 rad/s)	Gel firmness, elasticity, cohesiveness, hand-feel moldability	Power-law Dynamic: $G' = K'\omega^{n'}$; Complex Viscosity: $\eta^* = K^*\omega^{n^*-1}$	$R^2 \geq 0.96$ across 16 accessions	High shear degrades gels with low complex viscosity ($\eta^* <$ 600 Pa · s)	Amylose content positively correlates with G' ($r = 0.88$), driving gel elasticity.	Industrial starch gels, texture-modified food products
Arueya & Ojesanmi (2019)	Chemical hydroxypropylation; Paste heating	Clarity, smoothness, paste stability, reduced retrogradation	RVA pasting kinetics equations	$R^2 = 0.96$ for degree of substitution vs pasting temp	Chemical modification lowers gelatinization threshold to 61.9 °C	Substitution weakens internal hydrogen bonds, increasing paste clarity and freeze-thaw stability.	Frozen food sauces, modified starch gels, industrial thickeners

Table S4: Thermal Gelatinization (DSC), Pasting (RVA), and Macromolecular Drivers

Parameter / Category	**Sharma & Ghoshal (2026)**	**Akissoé et al. (2011)**	**Lee & Yoo (2011)**	**Colin-Henrion et al. (2007)**	**Ayetigbo et al. (2025)**	**Adeboyejo (Thesis)**	**Argaw et al. (2024)**	**Correa Mosquera et al. (2025)**	**Lozano et al. (2022)**
Primary Matrix / Cultivars	Kidney bean native & nano-starch	*D. rotundata* (Anago, Gnidou, Kokoro, Laboco, Morokoro) & *D. alata* (Florido)	Sweet potato starch (native & hydroxypropylated, MS: 0.042–0.153)	Pureed plant foods review (tomato, potato, sweet potato, guava, etc.)	19 yam genotypes (8 *D. rotundata*, 11 *D. alata*)	*Colocasia esculenta*, *Xanthosoma sagittifolium*, *Dioscorea dumetorum*	*D. bulbifera* (SC4), *D. cayenensis* (A), *D. rotundata* (AC-04), *D. cayenensis* complex (CH-21)	Yam (*Dioscorea* spp.) starch in pandebono, pandeyuca, panderos, and cookies	*D. rotundata* tuber mucilage in hydrocolloid systems (0.4% & 0.8% with CMC)
DSC Onset (T_o)	44.57 °C	66.6–72.2 °C	Native: 66.6±0.2 °C (HPSPS drops to 57.7 °C)	~57 °C (sweet potato purees)	—	White bitter: 88.67 °C; Yellow bitter: 77.51 °C; Cocoyam: 72.30–92.70 °C	*D. rotundata*: 47.23±1.89 °C; *D. cayenensis*: 59.13±2.85 °C	—	—
DSC Peak (T_p)	66.88 °C	66.6–75.8 °C	Native: 76.9±0.2 °C	50–70 °C (potato starches)	—	White bitter: 99.90 °C; Yellow bitter: 94.26 °C	*D. rotundata*: 71.70±0.00 °C; *D. cayenensis*: 81.03±1.35 °C	—	—

DSC Conclusion (T_c)	~80–85 °C	—	Native: 82.6±0.2 °C	—	—	White bitter: 113.85 °C; Yellow bitter: 95.84 °C	*D. rotundata*: 105.20±5.55 °C; *D. cayenensis*: 100.93±3.17 °C	Gelatinization temp cited at 84.2 °C	—
DSC Enthalpy (ΔH)	-12.798 J/g	9.0–12.8 J/g db (Florido: 12.8, Laboco: 9.7)	Native: 16.5±0.2 J/g (HPSPS: 12.3 J/g)	—	—	White bitter: 16.88 J/g; Yellow bitter: 9.26 J/g; Cocoyam: 10.28–18.45 J/g	*D. rotundata*: 2.30±0.27 J/g; *D. bulbifera*: 5.39±0.21 J/g	—	—
RVA Peak Viscosity (PV)	—	—	—	—	1895–3036 cP	Native white bitter: 366.70±1.82 RVU	*D. cayenensis*: 2923.66 cP; *D. rotundata*: 2117.00 cP	—	—
RVA Trough Viscosity (TV)	—	—	—	—	1622–2538 cP	—	1216.00–2055.66 cP	—	—
RVA Breakdown / Setback	—	—	—	—	—	Peak drops <4 RVU under acid hydrolysis	Breakdown: 163.00 cP (*rotundata*) to 1000.66 cP (*cayenensis*); Setback: 233.66–2097.66 cP	—	—
RVA Pasting Temp	—	—	—	—	75.9–79.2 °C (*D. rotundata*) vs. 77.6–81.1 °C (*D. alata*)	—	74.73±0.63 °C to 79.50±0.00 °C	—	—
Amylose Content	Native: 12.18±0.35 %	Soluble: 5.5–13.1 g/100 g db	Native: 24.6 g/100 g	Leached amylose increases stickiness	*D. rotundata*: 19.0–30.6%; *D. alata*: 20.0–25.5%	White bitter: 8.69%; Yellow bitter: 17.95%; Cocoyam: 10.11–37.08%	*D. cayenensis*: 23.53%; *D. rotundata*: 19.85%; *D. bulbifera*: 16.92%	24.08%	—
Other Starch Fractions / Drivers	Nano-starch amylose unmeasurable	Soluble starch: 22.3–46.6 g/100 g db; β-glucan: 0.04–0.10 mg/g db	—	—	Total Pectin: *D. rotundata* (771–1147 mg/100 g), *D. alata* (608–894	—	Total Starch: 52.97–64.63%; Soluble Sugar: 2.36–4.95%	—	—

					mg/100 g)				

Table S5: Predictive Regression Equations and Constitutive Flow Models for Tuber Starch and Gel Matrices

Paper / Source	Matrix / Species	Predictive Equation or Flow Model	Key Model Variables	Fit (R2 / P)
Ayetigbo et al. (2025)	Pounded Yam (*D. rotundata* & *D. alata*)	$\mathrm{EXT} = -9.587 - 0.778\,\mathrm{SOL_{flour}} + 0.0133\,\mathrm{PV_{flour}}$	EXT: Extensibility (mm) $\mathrm{SOL_{flour}}$: Flour solubility (%) $\mathrm{PV_{flour}}$: Peak viscosity (cP)	$R^2 = 0.828$ $P = 0.0051$
Ayetigbo et al. (2025)	Pounded Yam (*D. rotundata* & *D. alata*)	$\mathrm{EPF} = 1.032 - 0.0372\,\mathrm{SOL_{flour}} + 0.000856\,\mathrm{PV_{flour}} - 0.000162\,\mathrm{SOL_{flour}^{-1.696}}\,\mathrm{PV_{flour}^{-2544}}$	EPF: Peak Force / Firmness (N) $\mathrm{SOL_{flour}}$: Flour solubility (%) $\mathrm{PV_{flour}}$: Peak viscosity (cP)	$R^2 = 0.920$ $P = 0.0035$
Akissoé et al. (2011)	Cooked Yam (*D. rotundata* & *D. alata*)	$\mathrm{Firmness} = -6.18 + 0.41\,\mathrm{DM} - 0.14\,\mathrm{St} + 0.45\,\mathrm{Amy}$	Firmness: Extrusion firmness ($\mathrm{N/cm^2}$) DM: Dry matter (g/100 g wb) St: Soluble starch (g/100 g db) Amy: Amylose (g/100 g db)	$R^2 = 0.75$
Gumus & Wauton (2014)	Yam Starch Paste (*D. rotundata*, 40 °C)	$\tau = 26.36\,\gamma^{0.388}$	τ: Shear stress (Pa) γ: Shear rate ($\mathrm{s^{-1}}$) $K = 26.36\,\mathrm{Pa \cdot s^n}$, $n = 0.388$	$R^2 > 0.98$
Gumus & Wauton (2014)	Yam Starch Paste (*D. rotundata*, 50 °C)	$\tau = 25.47\,\gamma^{0.388}$	τ: Shear stress (Pa) γ: Shear rate ($\mathrm{s^{-1}}$) $K = 25.47\,\mathrm{Pa \cdot s^n}$, $n = 0.388$	$R^2 > 0.98$
Gumus & Wauton (2014)	Yam Starch Paste (*D. rotundata*, 60 °C)	$\tau = 24.77\,\gamma^{0.336}$	τ: Shear stress (Pa) γ: Shear rate ($\mathrm{s^{-1}}$) $K = 24.77\,\mathrm{Pa \cdot s^n}$, $n = 0.336$	$R^2 > 0.98$
Sharma & Ghoshal (2026)	Native Kidney Bean Starch Gel	$\tau = 3.96 + 0.0768\,\gamma^{0.762}$	τ_0: Yield stress (3.96 Pa) K: Consistency index ($0.0768\,\mathrm{Pa \cdot s^n}$) n: Flow behavior index (0.762)	$R^2 > 0.99$
Sharma & Ghoshal (2026)	Nano-Starch Gel	$\tau = 3.15 + 0.0153\,\gamma^{0.870}$	τ_0: Yield stress (3.15 Pa) K: Consistency index ($0.0153\,\mathrm{Pa \cdot s^n}$) n: Flow behavior index (0.870)	$R^2 > 0.99$
Lozano et al. (2022)	Pure Yam Mucilage (0.8% Concentration)	$\tau = 0.023\,\gamma^{0.900}$	K: Consistency index ($0.023\,\mathrm{Pa \cdot s^n}$) n: Flow behavior index (0.900)	$R^2 > 0.994$
Lozano et al. (2022)	Pure Yam Mucilage (0.8%) Oscillation Moduli	$G' = 1.629\,\omega^{2.068}, \quad G'' = 0.907\,\omega^{2.050}$	G': Storage modulus (Pa) G'': Loss modulus (Pa) ω: Angular frequency (rad/s)	$R^2 > 0.98$